English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, 2020/2021 Written exam, 27.01.2021

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (1 point) An urn contains balls numbered from 1 to 10 inclusive. Three balls are drawn without replacement. Define the following events concerning the balls drawn:

1. $A =$*the largest number is 6*;
2. $B =$*at least two balls have even numbers*;
3. $C =$*the sum of the numbers is divisible by 3.*

Find the probabilities of these events and determine whether any are independent. What is $\mathbb{P}(A|B)$? What is the expected number of even-numbered balls drawn if sampling is with replacement?

Problem 2. (1 point) Two instruments measure the concentration of dust particles in the air. Suppose the true concentration is x g/m^3. The reading of the first instrument is normally distributed with mean x and standard deviation (σ) $0.05x$. The second reading is also normally distributed, with mean x and standard deviation $0.1x$. Which instrument would you use? For each instrument, what is the probability of an error greater than $0.1x$?

Someone decides to use the arithmetic mean of the two readings. If the measurements are independent, what is the probability of an error greater than $0.1x$ with this procedure?

Problem 3. (1 point) Let ξ and η be independent random variables, $\xi \sim Exp(2)$ and $\eta \sim U(0,3)$, i.e.

$$f_\xi(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}; \quad f_\eta(x) = \begin{cases} \frac{1}{3} & \text{if } 0 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Find the correlation of ξ and η, $P(\xi < \eta)$ and the density of ξ/η.

Problem 4. (1 point) Let ξ and η have joint density

$$f_{\xi,\eta}(x,y) = \begin{cases} cx^3y^2 & \text{if } 0 < x, y < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find the constant c, the correlation of ξ and η, and $\mathbb{E}(\xi|\eta = 1)$.